Reinforcement Learning Course Project

Clinical Treatment Optimisation: Sepsis ICU Management

Master in Data Science & Advanced Analytics

Reinforcement Learning Course

Group F

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Academic Year 2025/2026

github.com/mariamiguel720/Reinforcement-Learning-Project-25-26

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1. Introduction

Sepsis is a life-threatening organ dysfunction caused by a dysregulated host response to infection, and one of the leading causes of in-hospital mortality in the ICU. Its treatment consists of a sequence of decisions: every few hours the clinician must adjust vasopressor and intravenous fluid doses based on the patient's physiology, without knowing how the patient will respond.

These properties: repeated decisions, stochastic transitions, and delayed sparse outcome signals, fit naturally into a Markov Decision Process, which has become the standard formalism for data-driven sepsis management since the work of Komorowski et al. [4]. The same approach is adopted here: the patient's clinical status at each four-hour window is the state s_t, the chosen combination of vasopressor and fluid is the action a_t, and the reward encodes survival at discharge. The Markov assumption, that s_t is a sufficient statistic of the patient's history, is not fully realistic in a clinical environment where many factors may affect the outcome. However, it is reasonable given the rich physiological summary captured by the 47 features, and it is the assumption that makes this problem tractable as a Reinforcement Learning task.

This project applies reinforcement learning to the ICU-Sepsis-v2 benchmark [5] under two configurations. Configuration A uses a discrete state representation, enabling tabular methods, Dyna-Q with Prioritized Sweeping and Q-Learning, augmented with potential-based reward shaping. Configuration B replaces the discrete state with a continuous 47-dimensional physiological observation and adds clinical wrappers that simulate real ICU conditions, requiring deep RL methods: DQN (with a Double DQN extension) and PPO. Both configurations are evaluated against a random-policy baseline and assessed not only on survival rate but on the clinical coherence of the learned treatment strategies.

2. Background

2.1. The ICU-Sepsis-v2 Environment

ICU-Sepsis-v2 [5] is a Gymnasium benchmark derived from the MIMIC-III clinical database following Komorowski et al. [4]. Real ICU trajectories meeting Sepsis-3 criteria were aggregated into four-hour decision windows and clustered into 714 clinical states; two absorbing terminal states (survived, died) bring the total to $|\mathcal{S}| = 716$. Transition tensor P and reward matrix R are estimated directly from these trajectories.

Both configurations share the same action space, reward function, transition dynamics, and discount factor, differing only in state representation: Configuration A exposes a discrete cluster index; Configuration B replaces it with the underlying 47-dimensional physiological feature vector.

Action Space. Actions discretise the joint vasopressor–fluid dose choice into a 5×5 grid (25 actions, $a = 5v + f$, $v, f \in \{0, \dots, 4\}$). Treatment intensity $\iota(a) = (v + f)/8 \in [0, 1]$ is used throughout the reward and policy analysis.

Reward Function. A sparse terminal reward (+1 survival, 0 death) is augmented with a per-step intensity penalty:

$$R(s, a, s') = R_{\text{terminal}}(s') - \lambda\, \iota(a), \qquad \lambda = 0.02, \tag{1}$$

where $R_{\text{terminal}}(s') = +1$ only at survival. A discount factor $\gamma = 1$ is adopted following [5], so $\mathbb{E}_\pi[G_0] \approx \Pr_\pi(\text{survival})$. Initial states are biased toward more severe patients.

2.1.1. Configuration A: Discrete State Space

The agent observes a cluster index $s \in \{0, \dots, 715\}$. The resulting Q-table has 17,900 entries, making tabular methods directly applicable, and the full empirical model is exposed for model-based planning.

2.1.2. Configuration B: Continuous Observation Space

The cluster index is replaced by its centre in physiological feature space, $\mathbf{o} \in \mathbb{R}^{47}$, requiring function approximation via neural networks. Three episodic wrappers mimic real ICU conditions: Gaussian observation noise (prob. 0.15, std 0.10), missing features (prob. 0.15, 4 features zeroed), and rare acute death regardless of treatment (prob. 0.01 per step). Training and evaluation both use this clinical environment.

2.2. Empirical Characterisation of the Environment

Random-policy rollouts over 1,000 episodes establish the performance floor. The transition tensor P is sparse (88% zero entries), making model-based planning tractable in Configuration A (see Table 4).

Configuration A achieves 69.3% survival under a random policy; the clean Config B baseline is 71.9%, consistent with both configurations sharing the same underlying MDP. The -6.6 pp drop to the clinical environment (65.3%) quantifies the combined wrapper cost (see Table 4).

3. Methodology

3.1. Experimental Setup

Both configurations share a common set of fixed settings. The initial state distribution is biased toward more severe patients via `sofa_bias` $= 5.0$, yielding a mean SOFA score of approximately 8.8 at episode start. A treatment intensity penalty of $\lambda = 0.02$ is applied at every step. All random seeds are fixed for reproducibility. Final evaluation uses 2,000 episodes for Configuration A and 1,000 episodes for Configuration B, with 95% confidence intervals reported throughout.

3.2. Configuration A: Discrete State Space

3.2.1. Reward Shaping

Because the reward is extremely sparse (only the terminal step carries a meaningful signal), potential-based reward shaping [2] is applied to both algorithms. The value function V_π is first computed via Policy Iteration on the full MDP, and the reward is then shaped as:

$$r'(s, a, s') = r(s, a, s') + \gamma\, V_\pi(s')(1 - \text{done}) - V_\pi(s) \tag{2}$$

This transformation converts the sparse terminal signal into dense per-step feedback proportional to the change in survival probability, without altering the optimal policy. V_π ranges from 0.00 to 0.98 across clinical states (mean 0.827), meaning the average patient has roughly 83% chance of surviving under optimal treatment. Policy Iteration itself is not evaluated as a featured algorithm; it serves solely to provide V_π for shaping and as a reward-optimal reference policy. Its survival rate of 78.8% reflects the dynamic programming solution to the penalised reward objective (which balances survival against treatment intensity), not an upper bound on survival rate, a distinction that explains why Dyna-Q+PS, which learns a different balance between the two objectives, can exceed it on this metric.

3.2.2. Dyna-Q with Prioritized Sweeping

Dyna-Q with Prioritized Sweeping is selected because Configuration A exposes the full transition matrix $P(s' \mid s, a)$, making model-based planning both feasible and exact. After each real environment step, k additional Q-value updates are performed by sampling state-action pairs from a priority queue ordered by absolute TD error and simulating next states from the known model. The Q-value update for both real and simulated steps is:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r' + \gamma \max_{a'} Q(s', a') - Q(s, a) \right] \tag{3}$$

This structure multiplies the effective number of Q-value updates per real episode by $(1 + k)$. When the priority queue empties, the remaining planning steps fall back to random sampling over clinical states to maintain exploration coverage. The ε-greedy exploration schedule decays from $\varepsilon = 1.0$ to a floor of 0.005, and the learning rate follows $\alpha_t = \max(\alpha_{\text{end}},\ \alpha_{\text{start}} \cdot \alpha_{\text{decay}}^t)$.

3.2.3. Q-Learning

Q-Learning applies the same update rule (Equation 3) but only on real transitions, with no planning steps. It is selected specifically to create a controlled comparison: both algorithms share identical update equations, identical reward shaping, and identical seeds, so any performance difference at equal episode budgets is attributable directly to the k planning steps that Dyna-Q+PS adds. Q-Learning also serves as the natural model-free baseline in any tabular RL setting, with proven convergence guarantees under Robbins-Monro conditions [3].

3.2.4. Hyperparameter Search

Each algorithm was first trained with hand-picked baseline parameters (60,000 episodes) and then optimised using Optuna with TPE sampling (30 trials $\times$ 60,000 episodes, evaluated over 300 episodes per trial). The final optimised run trains for 120,000 episodes and evaluates over 2,000 episodes (95% CI $\pm$1.8%).

The Optuna search space for α_{end} was constrained to [0.0001, 0.01] after preliminary experiments showed that wider ranges allowed α_{end} close to α_{start}, preventing the learning rate from decaying and destabilising Q-values during extended training. This is discussed further in the Limitations subsection. The final hyperparameters are reported in Table 5.

The most significant Optuna finding was $k = 95$ for Dyna-Q+PS (up from the baseline's $k = 20$). At this level, each real episode generates 96 Q-value updates; over 120,000 episodes, that totals 11.5 million updates compared to Q-Learning's 120,000.

3.3. Configuration B: Continuous Observation Space

The continuous 47-dimensional observation space rules out tabular methods and requires function approximation. Two algorithms representing complementary families of deep RL are selected: DQN (value-based, off-policy) and PPO (policy-gradient, on-policy).

3.3.1. DQN

DQN [8] is an off-policy value-based method that approximates $Q(s, a)$ with a neural network and stabilises training through an experience replay buffer and a target network. It is the natural extension of the Q-Learning logic used in Configuration A to a continuous observation space, providing a direct methodological bridge between the two configurations.

DQN is additionally evaluated with a Double DQN correction [10], which decouples action selection from action evaluation in the Bellman target to mitigate the well-documented overestimation bias of vanilla DQN. The motivation for this extension is empirical: the DQN learning curve showed a persistent positive trend at the end of the 100,000-episode budget, suggesting the agent had not yet converged and that a bias correction could yield further gains. Double DQN uses a slightly deeper network (`QNetworkV2`: 47→256→256→128→25) compared to DQN (47→256→256→25), adding a compression stage before the output. Optuna-tuned hyperparameters are otherwise shared, so performance differences reflect both the bias correction and the increased network capacity.

The DQN backbone maps the 47-dimensional observation through two hidden layers ($47 \to 256 \to 256$, LayerNorm + ReLU) to a $256 \to 25$ Q-value output head, trained with MSE loss on a TD target produced by a separate target network. LayerNorm handles the heterogeneous feature scales without requiring explicit input normalisation.

3.3.2. PPO

PPO [9] is an on-policy actor-critic method that directly optimises a parameterised policy $\pi_\theta(a \mid s)$ via a clipped surrogate objective. It contrasts sharply with DQN in both update mechanism (on-policy rollouts vs off-policy replay) and learning target (policy gradient vs Bellman backup), allowing a direct comparison of the two paradigms under identical environment, evaluation, and tuning conditions.

The PPO backbone maps the 47-dimensional observation through two hidden layers ($47 \to 256 \to 256$, Tanh, orthogonal initialisation with gain $\sqrt{2}$) shared by both heads: an actor ($256 \to 25$ categorical distribution over actions) and a critic ($256 \to 1$ value estimate). The network is trained with a clipped surrogate loss L^{CLIP} combined with a value loss and an entropy bonus. Orthogonal initialisation with tanh activations is the standard PPO setup and ensures stable gradient flow through the actor-critic backbone.

Both networks are kept deliberately small, as the sparse-reward environment and relatively low-dimensional state space mean additional capacity offers little benefit while amplifying variance across seeds.

3.3.3. Training and Tuning Protocol

The same three-stage protocol is applied to both algorithms, allowing direct comparison without confounding from methodological differences.

Phase 1 — Baseline run. A single training run with fixed, canonical hyperparameters is performed before any optimisation. This verifies implementation stability, establishes a rough return range to guide search bounds, and provides a reference against which the tuned agent is compared.

Phase 2 — Multi-seed Optuna search. Multi-seed Optuna search. Deep RL is known to exhibit substantial variance across random seeds even under identical hyperparameters [11], so tuning is performed across multiple seeds to avoid selecting configurations whose apparent performance reflects a single favourable initialisation. Hyperparameters are tuned with Optuna using a TPE sampler and a median pruner. Each trial is evaluated across three seeds $\{42, 123, 7\}$ and the objective is the mean evaluation return averaged across seeds. Inner pruning is enabled on the first two seeds so that clearly unpromising configurations are cut early. Each DQN trial trains for 10,000 episodes; each PPO trial runs for 100 to 200 updates (sampled from $\{100, 150, 200\}$), with the `MedianPruner` inactive for the first 75 updates to allow sufficient stabilisation before pruning decisions. A total of 20 trials were scheduled for DQN (17 completed) and 15 for PPO (9 completed), with the remainder pruned for poor early-stage performance. Due to computational constraints, only this phase applied multi-seed evaluation; subsequent stages used the single seed showing the best combination of peak performance and consistent learning dynamics.

Phase 3 — Final retrain and seed selection. With the best configuration identified by Optuna, each algorithm is retrained from scratch across the same three seeds at the full training budget (N=100,000 episodes for DQN and Double DQN, N=225 updates of 4,096 rollout steps for PPO). The best checkpoint seen during training is restored rather than the final weights, guarding against late-stage instability. The seed retained for final evaluation is selected by a qualitative criterion combining three indicators: highest best-eval value, positive learning-curve slope sustained throughout the budget, and lowest evaluation standard deviation. Best-eval alone is not used as the selection criterion, since a high peak followed by deterioration indicates an unstable run rather than a robust policy.

3.3.4. Hyperparameter Search Spaces

Table 6 lists the parameters tuned and the values held fixed for each algorithm. Search ranges were chosen around canonical defaults informed by the Phase 1 baseline run.

4. Evaluation

4.1. Configuration A

4.1.1. Random Baseline

A uniformly random policy was evaluated over 1,000 episodes on the Configuration A environment (`sofa_bias = 5.0`, $\lambda = 0.02$) to establish the performance floor that any learned policy must clearly exceed. The random policy achieves a **69.3% survival rate** (mean return 0.593, mean episode length 10.1 steps). The episode-length distribution is right-skewed, reflecting the fact that more severely ill patients, overrepresented by the SOFA bias, tend to reach terminal states faster. This baseline serves as the lower bound for all subsequent comparisons (see Figure 7).

4.1.2. Results

Table 1 summarises the final evaluation results for all Configuration A agents. Optimised runs are evaluated over 2,000 episodes (95% CI ±1.8%); baseline runs over 1,000 episodes (95% CI ±2.5%). The Agreement PI column measures the proportion of clinical states in which the greedy action of the learned policy matches the theoretically optimal action produced by Policy Iteration.

Table 1. Final evaluation results for Configuration A.

Algorithm	Survival	CI	vs PI	vs Random	Agree PI
PI (reference)	78.8%	±2.5%	—	+9.5 pp	100%
Dyna-Q+PS Baseline (60k)	79.6%	±2.5%	+0.8 pp	+10.3 pp	69.9%
Dyna-Q+PS Optuna (120k)	**81.1%**	±1.8%	+2.3 pp	+11.8 pp	**90.5%**
Q-Learning Baseline (60k)	76.2%	±2.6%	−2.6 pp	+6.9 pp	31.0%
Q-Learning Optuna (120k)	77.5%	±1.8%	−1.3 pp	+8.2 pp	33.1%
Random Baseline	69.3%	—	−9.5 pp	—	—

Dyna-Q+PS (Optuna) reaches **81.1% survival**, the highest of all agents, exceeding the PI reference by 2.3 pp and Q-Learning (Optuna) by 3.7 pp. That Dyna-Q+PS surpasses PI on survival rate is consistent with the Methodology note: PI optimises the penalised return objective, not raw survival, and the two agents arrive at different trade-offs between treatment intensity and patient outcomes. Optuna improved both algorithms: Dyna-Q+PS by +1.5 pp (79.6% to 81.1%) and Q-Learning by +1.2 pp (76.2% to 77.5%). Dyna-Q+PS (Optuna) also achieves the highest Agreement PI at 90.5%, compared to 33.1% for Q-Learning across both its baseline and optimised runs (see Figure 8).

4.1.3. Convergence and Exploration - Exploitation Balance

Both algorithms use ε-greedy exploration decaying from $\varepsilon = 1.0$ to a floor of 0.005. At the Optuna decay rates, ε reaches the floor at approximately episode 26,500 for Dyna-Q+PS ($\varepsilon_{\text{decay}} = 0.9998$) and episode 10,600 for Q-Learning ($\varepsilon_{\text{decay}} = 0.9995$). The fact that survival continues improving after ε reaches its floor reveals that gains come from ongoing Q-value refinement, but only for Dyna-Q+PS, whose α remains active throughout training. For Q-Learning, α is already frozen by episode 3,200 (as $\alpha_{\text{start}} = 0.047$ reaches $\alpha_{\text{end}} = 0.0037$ with $\alpha_{\text{decay}} = 0.9992$), so all post-floor improvement comes solely from reduced evaluation-time exploration noise. This distinction is visible in the learning curves: Dyna-Q+PS shows sustained improvement throughout training, whereas Q-Learning's curve flattens once ε stabilises.

4.1.4. Learning Curves

The learning curves plot the rolling-average return (window = 500 episodes) against training episodes on a fixed y-axis $[0.55, 0.92]$, enabling direct visual comparison across all algorithms and training budgets.

Dyna-Q+PS Baseline (60k episodes). The agent learns rapidly in the first 20,000 episodes, after which the curve stabilises around 0.75–0.80. The positive trend of +0.018 return per 10k episodes at episode 60,000 confirms the algorithm is still improving when training ends, indicating that the 60,000-episode budget is insufficient for full convergence. The final smoothed return is 0.787 (PI reference: 0.751) (see Figure 9).

Dyna-Q+PS Optuna (120k episodes). Optuna selected $k = 95$ planning steps and a slow alpha decay ($\alpha_{\text{decay}} = 0.99996$), keeping the learning rate above its floor (0.00088) until approximately episode 141,000. Q-values therefore continue to be refined throughout the entire training run. The trend at episode 120,000 is +0.004 per 10k episodes: the agent is approaching convergence but has not fully plateaued. The final smoothed return is 0.767 (see Figure 1).

Q-Learning Baseline (60k episodes). The learning curve shows a steady positive trend throughout (+0.015 per 10k episodes), confirming that 60,000 episodes is far below the convergence requirement for a model-free method without planning steps. The final smoothed return is 0.725 (see Figure 10).

Q-Learning Optuna (120k episodes). The Optuna search selected a fast alpha decay ($\alpha_{\text{decay}} = 0.9992$), driving the learning rate to its floor of 0.0037 by episode 3,200, leaving it essentially frozen for the remaining 117,000 episodes. After that point, improvement is driven solely by reduced exploration noise as ε approaches its floor, not by continued Q-value refinement. The trend at episode 120,000 is +0.004 per 10k episodes (see Figure 2).

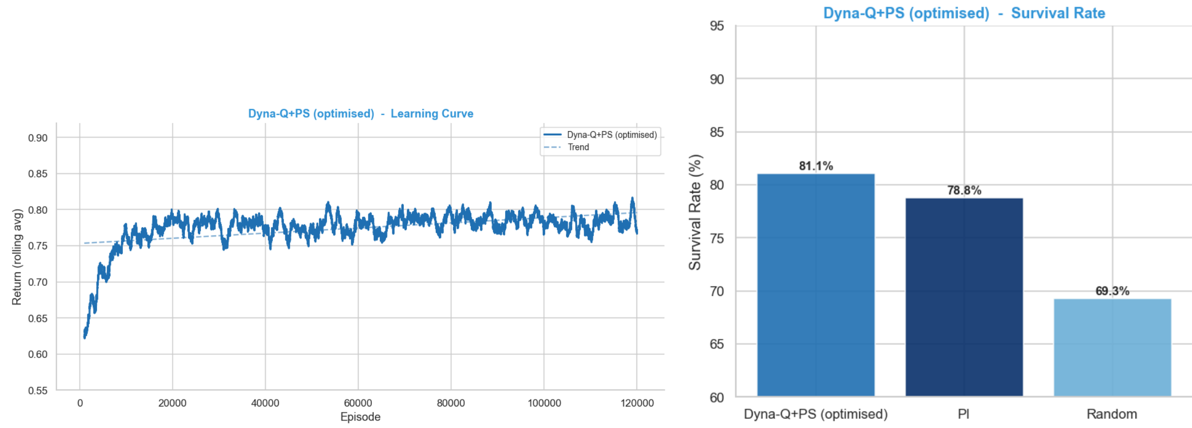

Figure 1. Dyna-Q+PS Optimised: learning curve (left) and survival rate (right).

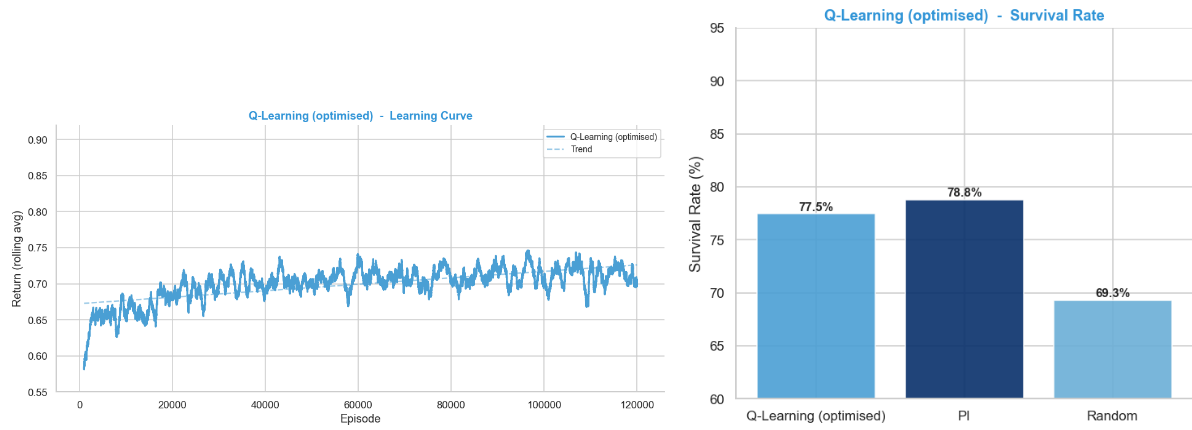

Figure 2. Q-Learning Optimised: learning curve (left) and survival rate (right).

The Optuna optimisation histories for both algorithms are shown in Figure 3. For Dyna-Q+PS, the best trial (82.7%) was found at trial 11 of 30, with no subsequent trial improving on this configuration, suggesting adequate coverage of the hyperparameter space. For Q-Learning, the best trial (76.0%) was found at trial 7 of 30 and performance stabilised thereafter.

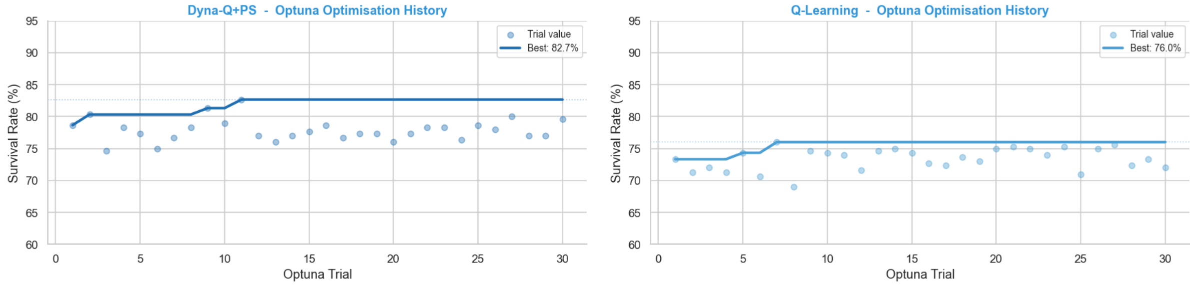

Figure 3. Optuna optimisation history for Dyna-Q+PS (left) and Q-Learning (right). Each point is a trial survival rate; the solid line tracks the cumulative best.

4.1.5. Algorithm Comparison

Three findings summarise the comparison between the two Configuration A algorithms.

Planning advantage. The 3.7 pp survival gap between Dyna-Q+PS (81.1%) and Q-Learning (77.5%) at matched episode budgets is entirely attributable to the $k = 95$ planning steps (Section 3.2.2). In a sepsis setting where the reward signal is sparse and delayed, access to the empirical transition model allows Dyna-Q+PS to extract far more learning signal per real interaction than a model-free agent can. From a clinical perspective, this translates directly into fewer suboptimal treatment decisions: Dyna-Q+PS matches the provably optimal action in 90.5% of patient states, against 33.1% for Q-Learning, meaning Q-Learning recommends a suboptimal vasopressor or fluid dose in roughly two out of every three clinical states.

Algorithm choice over hyperparameter tuning. Even the untuned Dyna-Q+PS baseline (79.6%) outperforms optimised Q-Learning (77.5%), confirming that structural algorithmic differences dominate tuning effects in this environment. For clinical deployment this matters: it implies that investing in the right algorithm for the problem structure, one that exploits the available transition model, yields more reliable improvements than extensive hyperparameter search on a weaker algorithm.

Policy quality and therapeutic futility. Agreement PI is the most clinically meaningful metric in this setting, since two policies can achieve similar survival rates while recommending fundamentally different treatments. Optuna improved Dyna-Q+PS's Agreement PI from 69.9% to 90.5% but left Q-Learning essentially unchanged (31.0% to 33.1%), confirming that Q-Learning's policy structure never converged toward the optimal regardless of training length. The clinical consequence of this divergence is examined in Section 5: Dyna-Q+PS reproduces a non-obvious therapeutic futility strategy, allocating maximum treatment intensity to moderately severe patients rather than the most critical, that Q-Learning consistently fails to discover, instead treating monotonically with severity and incurring unnecessary penalties.

4.2. Configuration B: Continuous Observation Space

4.2.1. Random Baseline

A uniformly random policy was evaluated over 1,000 episodes on both the base environment and the clinical environment to establish the performance floor for Configuration B. On the clean base environment the random policy achieves 71.9% survival with a mean episode length of 10.0 steps (see Figure 11). On the clinical environment, which adds the three episodic wrappers, survival drops to 65.3% with a mean return of 0.5644, quantifying the difficulty added by the wrappers. All trained agents are compared against the clinical environment baseline.

4.2.2. Results

Table 2 summarises the final evaluation results for all Config B agents, evaluated over 1,000 episodes on the clinical environment.

Table 2. Final evaluation results for Configuration B. All agents compared against the clinical environment random baseline (65.3% survival, mean return 0.5644).

Algorithm	Survival	Return	vs Random (Survival)
Random baseline (clinical)	65.3%	0.5644	–
DQN Run 1 (baseline)	65.9%	0.5527	+0.6 pp
DQN Tuned (seed 123)	67.3%	0.5948	+2.0 pp
Double DQN (seed 42)	67.0%	0.5722	+1.7 pp
PPO Run 1 (baseline)	68.0%	0.6192	+2.7 pp
PPO Tuned (seed 7)	**71.0%**	**0.6787**	**+5.7 pp**

PPO Tuned achieves the best overall performance at 71.0% survival, outperforming the random baseline by 5.7 pp and the best DQN variant by 3.7 pp. All trained agents exceed the random baseline in survival rate, confirming genuine learning across both algorithm families. DQN Run 1 is the sole exception in mean return (0.5527 vs 0.5644 for random), reflecting its higher treatment intensity rather than worse patient outcomes.

4.2.3. Convergence Analysis

The multi-seed training results reveal substantial run-to-run variance across all algorithms, underscoring the importance of seed selection beyond peak performance alone.

The Optuna search for DQN converged at trial 0, with no subsequent trial able to improve the best configuration. The search consistently favours a very low learning rate ($\approx 2.2 \times 10^{-5}$) and a slow epsilon decay (≈ 0.9993) (see Figure 16).

DQN across three seeds shows best-eval values ranging from 0.6933 (seed 123) to 0.7137 (seed 42). Seed 42 achieves the highest best-eval (0.7137) and the strongest positive growth (+0.0966), but also the highest variance (0.0566), reflecting a more unstable learning trajectory. Seed 123 was selected for its sustained positive growth (+0.0193) and consistent upward slope throughout training (see Figure 12).

Double DQN across three seeds shows best-eval values ranging from 0.6980 (seed 123) to 0.7087 (seed 42). Seed 42 is selected as it combines the highest best-eval with the strongest positive growth (+0.1288) and steepest slope (4.34×10^{-7}) (see Figure 13).

The PPO Optuna search completed 9 of 15 scheduled trials, consistently favouring very low learning rates ($\text{lr} < 10^{-4}$) and the maximum rollout length ($n\_steps = 4096$) (see Figure 18).

PPO across three seeds shows the highest best-eval values of all Config B agents, ranging from 0.6475 (seed 123) to 0.7292 (seed 7). All three seeds show positive growth and positive slope, confirming that PPO learns reliably across different initialisations. Seed 7 is selected for its highest best-eval and consistent upward trend (see Figure 14).

4.2.4. Learning Curves

The learning curves are presented first for the baseline runs, establishing whether each algorithm shows initial learning capacity, and then for the tuned models, confirming sustained improvement over the full training budget.

DQN baseline (Run 1) trained for 50,000 episodes shows high variance throughout with no clear upward trend, confirmed by a near-zero slope of -1.05×10^{-8}. The agent plateaued early with the default hyperparameters, motivating the Optuna search (see Figure 15).

DQN Tuned (seed 123) trained for 100,000 episodes shows a clear positive trend (slope $= 3.87 \times 10^{-7}$), with returns stabilising in the 0.58–0.63 range by the end of training (see Figure 4).

Double DQN (seed 42) trained for 100,000 episodes shows a stronger positive trend (slope $= 4.34 \times 10^{-7}$) with returns reaching the 0.60–0.64 range by the end of training (see Figure 5).

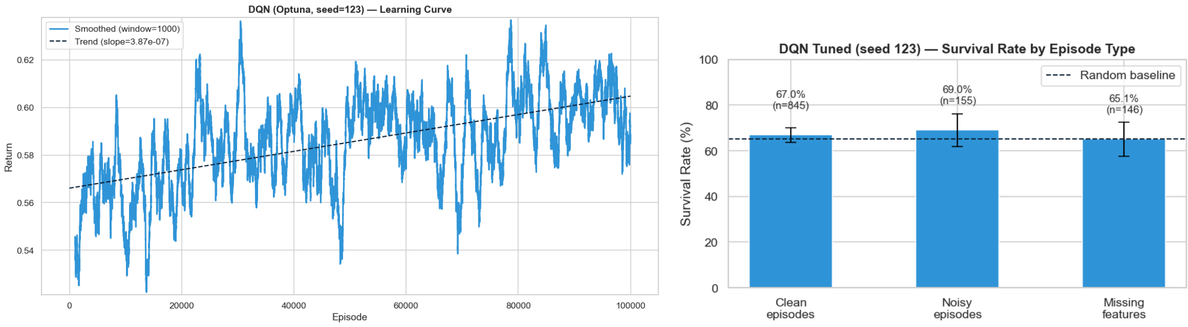

Figure 4. DQN Tuned (seed 123): learning curve (left) and survival rate by episode type (right).

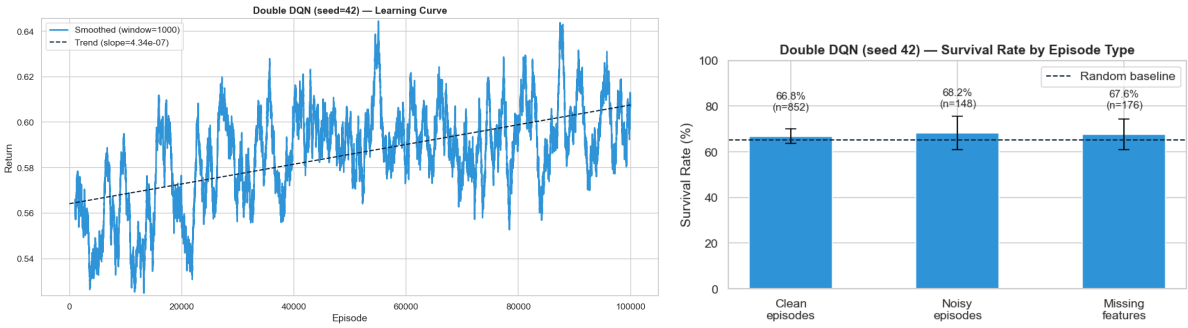

Figure 5. Double DQN (seed 42): learning curve (left) and survival rate by episode type (right).

PPO baseline (Run 1) trained for 150 updates (≈70,000 episodes) already shows a positive trend (slope $= 5.01 \times 10^{-7}$) with the default hyperparameters, indicating that PPO is more stable than DQN under canonical settings in this environment (see Figure 17).

PPO Tuned (seed 7) trained for 225 updates (≈100,000 episodes) shows the strongest and most consistent upward trend of all agents (slope $= 6.36 \times 10^{-7}$), with returns reaching the 0.62–0.65 range and no signs of plateau.

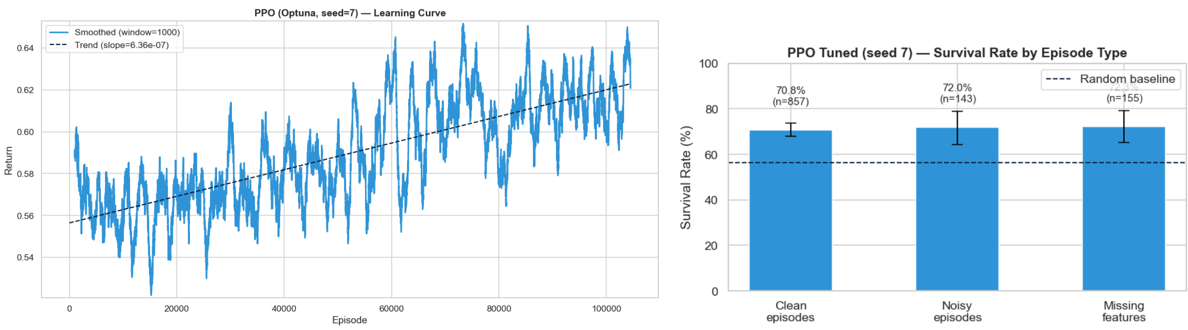

Figure 6. PPO Tuned (seed 7): learning curve (left) and survival rate by episode type (right).

4.2.5. Algorithm Comparison

Three findings summarise the comparison between Config B agents.

PPO outperforms DQN variants. PPO Tuned achieves 71.0% survival against 67.3% for DQN Tuned and 67.0% for Double DQN, a gap of 3.7 pp. The learning curves (Section 4.2.4) confirm this ranking, with PPO Tuned showing the steepest upward trend across all agents. The on-policy nature of PPO, collecting fresh experience at every update, appears better suited to this sparse-reward environment. From a clinical standpoint, however, this same property is PPO's main deployment liability: on-policy training requires data collected under the current policy, which in a real ICU would mean exposing patients to an unvalidated controller. DQN's off-policy formulation, although weaker here, would allow training directly on historical MIMIC-III trajectories without any prospective patient exposure, an advantage that often outweighs a 3.7 pp simulator gap.

Double DQN shows faster learning than vanilla DQN. Despite near-identical final survival rates (67.0% vs 67.3%), the Double DQN learning curve shows a consistently steeper slope throughout training, suggesting more efficient Q-value updates. Clinically, the case for Double DQN is stronger than the survival numbers suggest: vanilla DQN's overestimation bias systematically inflates the apparent value of aggressive vasopressor and fluid combinations, biasing recommendations toward more interventionist policies. In an ICU, where over-treatment carries its own mortality cost, a bias-corrected estimator is the safer default regardless of matched evaluation survival.

Hyperparameter tuning matters more for PPO than for DQN. PPO improved by 3.0 pp from baseline to tuned (68.0% to 71.0%), while DQN improved by only 1.4 pp (65.9% to 67.3%). PPO's greater sensitivity to hyperparameters implies a higher validation burden before clinical deployment: a poorly tuned PPO agent performs no better than a random policy, whereas DQN degrades more gracefully. In resource-constrained settings where extensive tuning is not feasible, this robustness gap may favour DQN despite its lower peak performance.

4.3. Overall Comparison

Table 3 consolidates the best agent from each algorithm family across both configurations. The headline survival rates are not directly comparable, since the two configurations differ in observation space, in the presence of clinical wrappers, and in the random-policy baseline (69.3% for A, 65.3% for B). The most meaningful cross-configuration metric is the lift each agent achieves over its own random baseline.

Table 3. Best agent per algorithm family across both configurations.

Algorithm	Config	Survival	Return	vs Random (Survival)
Dyna-Q+PS	A	81.1%	0.767	+11.8 pp
Q-Learning	A	77.5%	0.696	+8.2 pp
PPO	B	71.0%	0.679	+5.7 pp
DQN	B	67.3%	0.595	+2.0 pp

The within-configuration pattern is consistent: the family that extracts more updates per environment interaction wins: Dyna-Q+PS over Q-Learning in A by 3.7 pp, PPO over DQN in B by 3.7 pp. Across configurations, the larger lift in A reflects two structural advantages absent in B: access to the exact empirical transition model for planning, and the absence of the acute-event wrapper, which alone caps attainable survival by approximately 5 pp regardless of policy quality. Dyna-Q+PS captures essentially all available improvement in A, exceeding the Policy Iteration reference by 2.3 pp; no Config B agent recovers the 6.6 pp lost to the clinical wrappers, with PPO closing the gap further than DQN.

These structural differences reflect a fundamental tension in applying RL to sepsis management. Configuration A represents an idealised setting: the agent has access to a compact discrete state space and the full empirical transition model, conditions that do not hold in a real ICU. Configuration B is the harder and more realistic scenario, continuous physiological observations, missing data, sensor noise, and irreducible patient deterioration, and it is there that the clinical deployment questions become concrete. The 5 pp survival cost of the acute wrapper cannot be recovered by any policy, since it reflects stochastic patient deterioration that no treatment decision can prevent. The remaining gap to Config A is attributable to the loss of the transition model, and PPO's 5.7 pp lift over random suggests that meaningful policy improvement is achievable even under these constraints. Whether that improvement translates to a real ICU depends on factors beyond algorithm choice, data quality, state representation, and the ethical and regulatory requirements for deploying an automated treatment controller.

5. Creative Extension

5.1. Configuration A: Clinical Decision Support Analysis

The Config A creative extension moves beyond algorithm evaluation metrics to ask the question that matters most in a clinical setting: do the agents' decisions make clinical sense, and what physiological reasoning underlies them? Three analyses use SHAP (SHapley Additive exPlanations) and policy analysis to interpret learned behaviour in terms of the 47 physiological variables shared with Config B. Each analysis is designed to answer a concrete clinical question.

5.1.1. V_π as a Clinical Acuity Score

Clinical question: has the agent learned a meaningful measure of patient severity?

$V_\pi(s) = \Pr(\text{patient in state } s \text{ survives under optimal treatment})$ is a data-driven acuity score derived entirely from patient outcomes rather than clinical rules. To evaluate whether it captures clinically meaningful severity, it is compared to the SOFA score, the gold-standard severity index used in ICUs worldwide.

A GradientBoosting surrogate model is fitted on the 714 clinical states (47 features $\to V_\pi$) with $R^2 = 0.9995$, confirming an excellent fit for SHAP attribution. The correlation between SOFA and V_π is $r = -0.485$: higher SOFA (more severe illness) is associated with lower survival probability under optimal treatment, as expected. The moderate rather than strong correlation reflects the fact that V_π captures multi-organ complexity beyond what a single scalar severity index encodes.

The top SHAP features driving V_π predictions are ALT, WBC, Insulin, SpO2, and RespRate (all with negative associations except RespRate). Elevated ALT and WBC indicate liver dysfunction and systemic inflammation respectively, two hallmarks of sepsis-induced organ failure. Low SpO2 signals respiratory compromise. Together these features are consistent with the known pathophysiology of sepsis progression, suggesting the agent has implicitly learned a clinically coherent model of patient prognosis from outcome data alone.

5.1.2. Treatment Strategy by Patient Severity

Clinical question: does the agent learn to escalate treatment for sicker patients?

A rational ICU agent should recommend more aggressive vasopressor and fluid therapy for states with lower V_π (worse prognosis). The mean treatment intensity of PI, Dyna-Q+PS, and Q-Learning is compared across V_π quartiles (Q1 = most critical, Q4 = most stable), (see Table 7).

PI and Dyna-Q+PS display a non-monotonic pattern: treatment intensity peaks at Q2 Severe (0.210 and 0.211) rather than Q1 Critical (0.148 and 0.166). This **therapeutic futility** strategy emerges from the per-step penalty $\lambda = 0.02$: for patients in Q1 Critical, survival probability is so low that aggressive treatment generates more penalty than expected survival benefit, making a more conservative approach the rational choice. Q4 Stable patients receive the least intervention (0.108 and 0.111), consistent with the no-treatment-when-unnecessary logic the penalty incentivises.

Q-Learning fails to learn this pattern entirely. Its intensities are uniformly high across all quartiles (0.320–0.403) with virtually no differentiation by severity, reflecting a policy that has not converged to the optimal treatment logic. Overtreatment of Q1 Critical patients incurs unnecessary penalties that partially explain Q-Learning's lower survival rate relative to Dyna-Q+PS.

5.1.3. Comparison of Learned Feature Importance

Clinical question: do algorithms that agree more with the optimal policy also weight the same clinical features?

Dyna-Q+PS achieves 90.5% policy agreement with PI yet its SHAP feature importance correlation with V_π is only $r = 0.016$. Q-Learning, despite its 33.1% policy agreement, shows a higher feature importance correlation of $r = 0.076$ (surrogate R^2: 0.9957 and 0.9964 respectively).

This apparent paradox reveals an important distinction between *policy agreement* and *value function structure*. Two agents can recommend the same treatment while "reasoning" about different physiological variables. Dyna-Q+PS arrives at PI-matching decisions through a learning process driven by transition chains across the state space; its $k = 95$ planning steps propagate value information through sequences of states rather than fitting individual feature correlations. Q-Learning, by contrast, updates Q-values only from individual transitions, producing a value function that more closely mirrors the marginal feature-outcome correlations present in V_π, yet still recommends suboptimal actions in two-thirds of states.

The clinical implication is that feature importance alone is an insufficient proxy for policy quality: an agent can weight the right clinical variables and still fail to derive the correct treatment strategy, while an agent with an apparently different internal representation can make near-optimal decisions. Evaluation of RL-based clinical decision support systems should therefore prioritise direct policy comparison against a clinical reference, as Agreement PI does here, rather than interpretability metrics alone.

5.2. Configuration B: Wrapper Ablation Study

The Config B clinical environment combines three wrappers: episodic Gaussian noise, episodic missing features, and rare acute deterioration events. Their joint cost is known (-6.6 pp random-policy survival vs the clean base environment), but the individual contribution of each remains unattributed. This matters for deployment: sensor noise and missing data can be addressed by clinical engineering, whereas stochastic deterioration cannot. All $2^3 = 8$ wrapper combinations are therefore ablated to isolate where the cost actually lies.

The best PPO configuration is retrained from scratch in each of the eight conditions for 200 updates, across seeds $\{42, 123, 7\}$. Each policy is evaluated on the same wrapper combination it was trained on. Mean survival across seeds is reported in Table 8.

The **Acute** wrapper alone causes the largest single drop (-5.4 pp), accounting for essentially all of the degradation seen in the full clinical environment. Irreducibly stochastic deaths inject noise into the reward signal that no policy can remove. Episodic Gaussian perturbations (**Noisy**) produce no measurable degradation ($+0.6$ pp): PPO's stochastic policy and the redundancy in the 47-dimensional observation absorb feature-level noise of this magnitude. The **Missing** wrapper, zeroing four features in 15% of episodes, costs only 0.8 pp, consistent with the high correlation among clinical variables, which lets the agent infer missing information from the remaining observations.

A purely additive model predicts ≈62% for the full clinical condition. The observed 64.0% exceeds both double-wrapper combinations involving acute events (63.1% and 62.4%), suggesting that simultaneous exposure to all three stressors regularises the policy more than any pairwise subset. The components that clinical engineering can directly improve, sensors and missing lab values, contribute almost nothing to performance loss; the component it cannot remove accounts for essentially all of it (see Figure 19).

6. Conclusion

6.1. Summary of Findings

Across both configurations, the algorithm that extracts more learning signal per environment interaction consistently outperforms its counterpart: Dyna-Q+PS over Q-Learning in Config A by 3.7 pp (81.1% vs 77.5%), and PPO over DQN in Config B by 3.7 pp (71.0% vs 67.3%). Both gaps share the same structural explanation, more updates per real interaction, but arise from different mechanisms: model-based planning in Config A, and on-policy fresh experience in Config B. Across configurations, Config A achieves a larger lift over its random baseline (+11.8 pp vs +5.7 pp for PPO) due to two structural advantages absent in Config B: access to the exact empirical transition model for planning, and the absence of the acute-event wrapper, which alone caps attainable survival by approximately 5 pp regardless of policy quality.

In Config A, the performance gap between Dyna-Q+PS and Q-Learning extends beyond survival rates. Dyna-Q+PS matches the provably optimal treatment in 90.5% of patient states against 33.1% for Q-Learning, a 57 pp difference in policy quality. The most clinically significant finding is the therapeutic futility pattern: both PI and Dyna-Q+PS allocate maximum treatment intensity to Q2 Severe patients rather than Q1 Critical patients. This emerges from the per-step penalty ($\lambda = 0.02$), which makes aggressive treatment of near-certain non-survivors net-negative. Q-Learning fails to discover this pattern, treating monotonically with severity and incurring unnecessary penalties. The SHAP analysis surfaces a further distinction: despite its 90.5% policy agreement with PI, Dyna-Q+PS shows a lower feature importance correlation with V_π ($r = 0.016$) than Q-Learning ($r = 0.076$). Policy agreement and value function structure are different things: an agent can weight different physiological variables and still arrive at near-optimal treatment decisions, while an agent that mirrors the outcome-feature correlations more closely can still recommend the wrong action in two-thirds of states. This finding cautions against using interpretability metrics alone as proxies for policy quality in clinical decision support systems.

In Config B, PPO's on-policy nature proves better suited to the sparse-reward environment, but carries a deployment liability: on-policy training requires data collected under the current policy, which in a real ICU would mean exposing patients to an unvalidated controller. DQN's off-policy formulation, though weaker in simulation, would allow training directly on historical MIMIC-III trajectories without prospective patient exposure. Double DQN's bias correction provides an additional clinical argument: vanilla DQN's overestimation bias systematically inflates the apparent value of aggressive vasopressor and fluid combinations, biasing recommendations toward over-treatment. In an ICU where over-treatment carries its own mortality cost, a bias-corrected estimator is the safer default regardless of matched survival rates. The wrapper ablation study reveals that the dominant obstacle to RL-based sepsis management is not data quality but irreducible stochasticity: the acute event wrapper alone accounts for essentially all performance degradation (−5.5 pp), while sensor noise and missing lab values contribute almost nothing.

6.2. Limitations

The fixed environment configuration (`sofa_bias` $= 5.0$, $\lambda = 0.02$) constrains the generalisability of the futility finding; a different λ would shift the threshold where aggressive treatment becomes net-negative. The empirical MDP derived from MIMIC-III encodes historical physician behaviour as transition dynamics, conflating correlation with causation and limiting causal inference about treatment effects. In Config B, final evaluation uses a single seed per algorithm; while the multi-seed Optuna search mitigates selection bias, reported survival rates may vary across initialisations. Performance estimates also reflect a fixed patient cohort distribution; out-of-distribution generalisation to ICUs with different patient demographics or care protocols remains untested. PPO's learning curve shows a positive trend at the end of the 225-update budget, indicating the agent had not fully converged. The SHAP analyses use surrogate models (all with $R^2 > 0.99$) rather than direct Q-table interpretation, which is approximate.

6.3. Potential Improvements

Four improvements could strengthen the results. First, constraining the Q-Learning α_{decay} search range to [0.99995, 0.99999] would exclude fast-decay configurations and prevent the learning rate from freezing within the first 3,200 episodes of a 120,000-episode run. Second, extending Dyna-Q+PS training to 200,000 episodes would allow full convergence, as the current trend of +0.004 per 10k episodes at episode 120,000 indicates residual learning capacity. Third, extending PPO training beyond 225 updates is similarly motivated by the persistent positive slope at the end of training. Fourth, running all final evaluations across multiple seeds and reporting mean $\pm$ standard deviation would provide a more robust performance estimate that accounts for training stochasticity in addition to evaluation variance.

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7. Appendix

Table 4. Random policy performance across configurations.

Configuration	Survival	Return	Length
A (discrete)	69.3%	0.593	10.1
B clean	71.9%	0.622	10.0
B clinical	65.3%	0.561	≈ 10

Table 5. Final hyperparameters for Dyna-Q+PS and Q-Learning (baseline and Optuna-optimised).

Parameter	DQ baseline	DQ Optuna	QL baseline	QL Optuna
α_{start}	0.1	0.249	0.1	0.047
α_{end}	0.005	0.00088	0.005	0.0037
α_{decay}	0.9999	0.99996	0.9999	0.9992
k	20	**95**	–	–
θ_{PS}	0.01	0.092	–	–
$\varepsilon_{\text{decay}}$	0.9999	0.9998	0.9999	0.9995
Episodes	60,000	120,000	60,000	120,000

Table 6. Optuna search spaces and fixed hyperparameters for Config B agents.

DQN

Parameter	Range / value
Tuned	
lr	$[10^{-5}, 10^{-2}]$ log
batch_size	$\{32, 64, 128, 256\}$
eps_decay	$[0.9990, 0.9995]$
target_update	$\{25, 50, 100, 200\}$
buffer_capacity	$\{50, 100, 200\}$ k
Fixed	
eps_start	1.0
eps_end	0.05
gamma	1.0

PPO

Parameter	Range / value
Tuned	
lr	$[10^{-5}, 10^{-3}]$ log
ent_coef	$[5{\cdot}10^{-3}, 10^{-1}]$ log
n_updates	$\{100, 150, 200\}$
n_steps	$\{2048, 4096\}$
Fixed	
clip	0.2
gae_lam	0.95
batch_size	256
vf_coef	0.5
max_grad	0.5
gamma	1.0

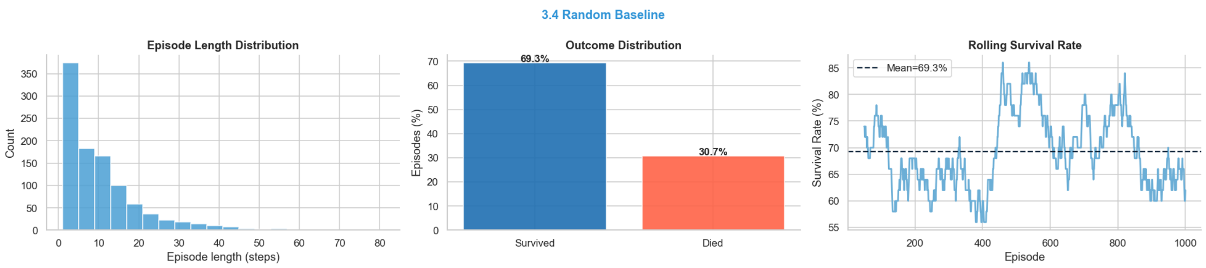

Figure 7. Random policy performance on Configuration A: episode length distribution (left), outcome distribution (centre), and rolling survival rate (right).

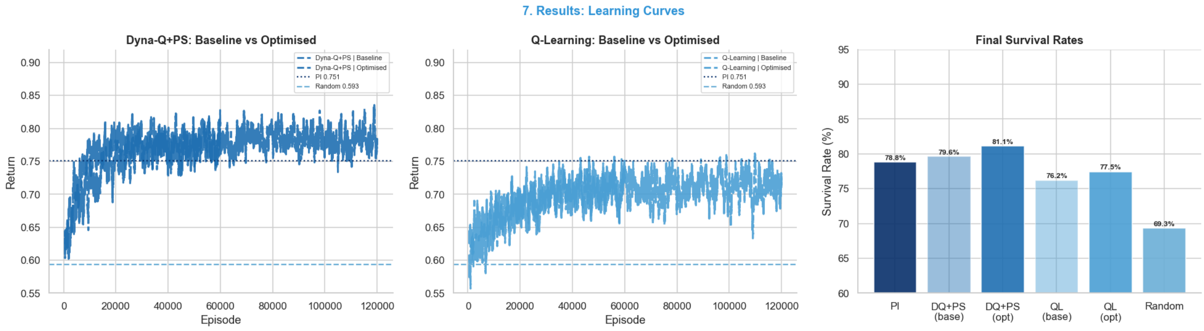

Figure 8. Configuration A summary: learning curves for both algorithms (left panels) and final survival rates compared against the PI reference and random baseline (right panel).

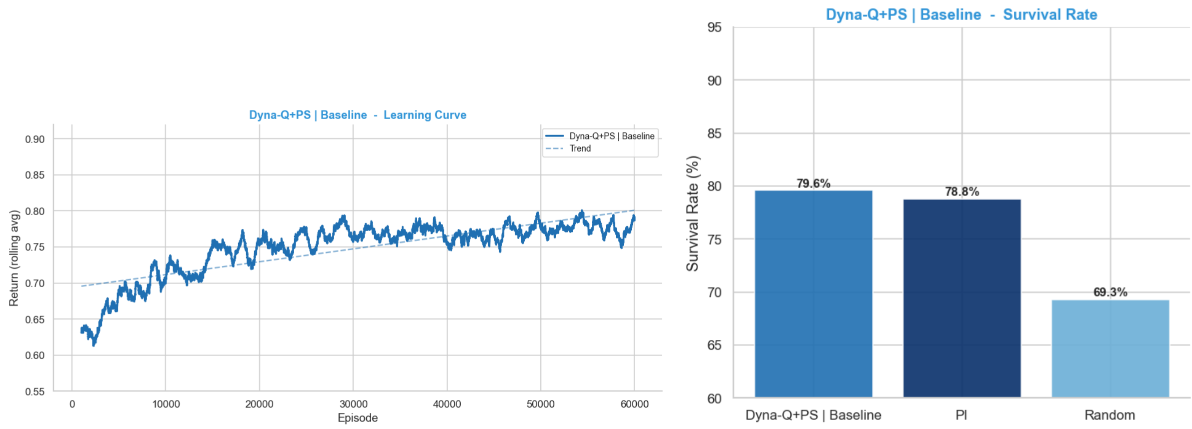

Figure 9. Dyna-Q+PS Baseline: learning curve (left) and survival rate (right).

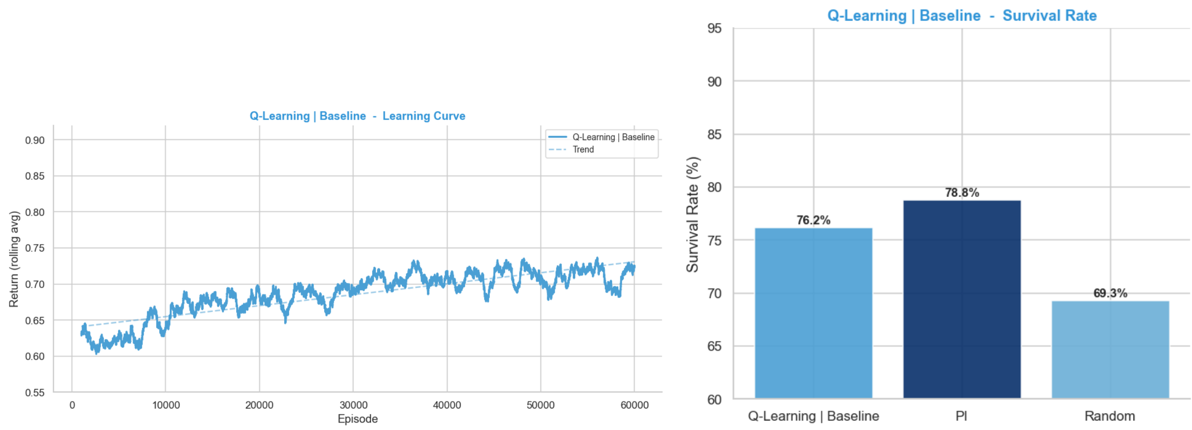

Figure 10. Q-Learning Baseline: learning curve (left) and survival rate (right).

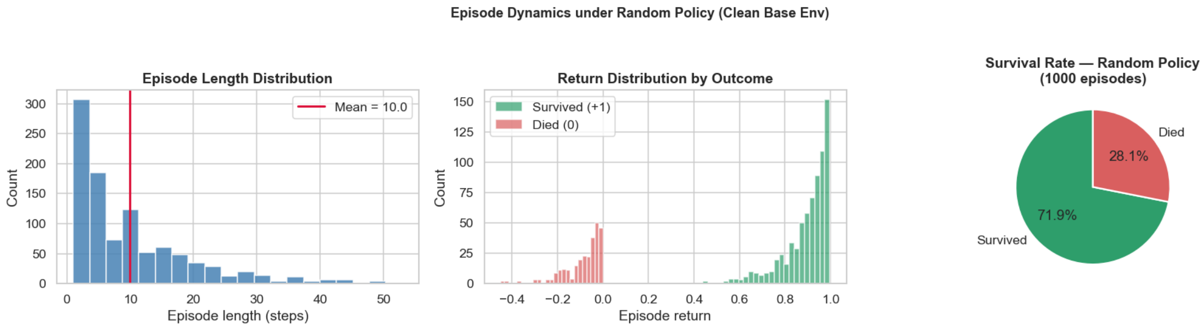

Figure 11. Episode dynamics under a random policy on the clean base environment (Config B). Left: episode length distribution. Centre: return distribution by outcome. Right: overall survival rate.

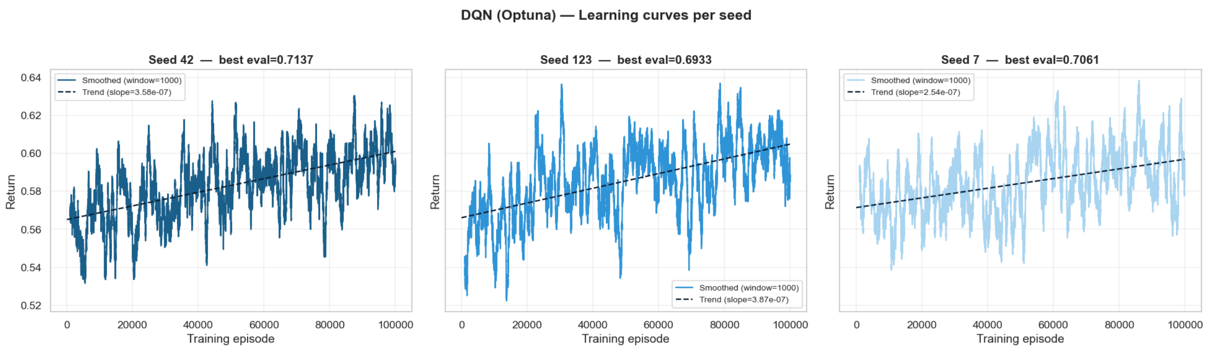

Figure 12. DQN (Optuna) learning curves across three seeds.

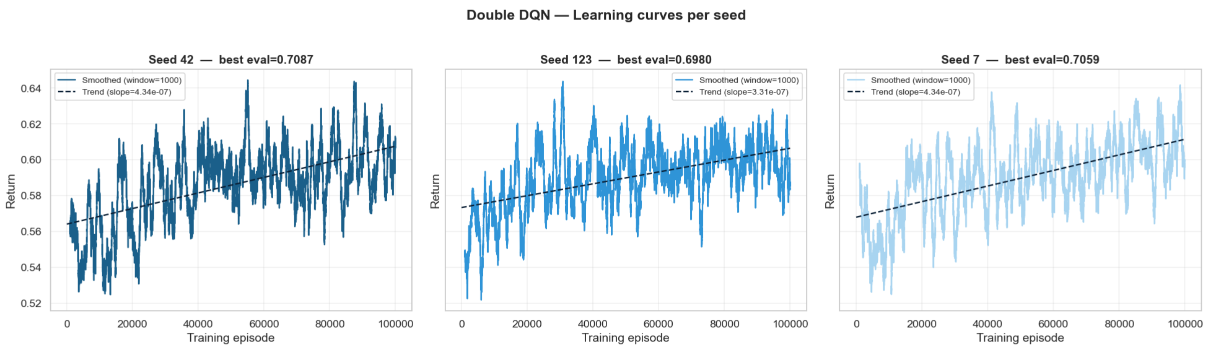

Figure 13. Double DQN learning curves across three seeds.

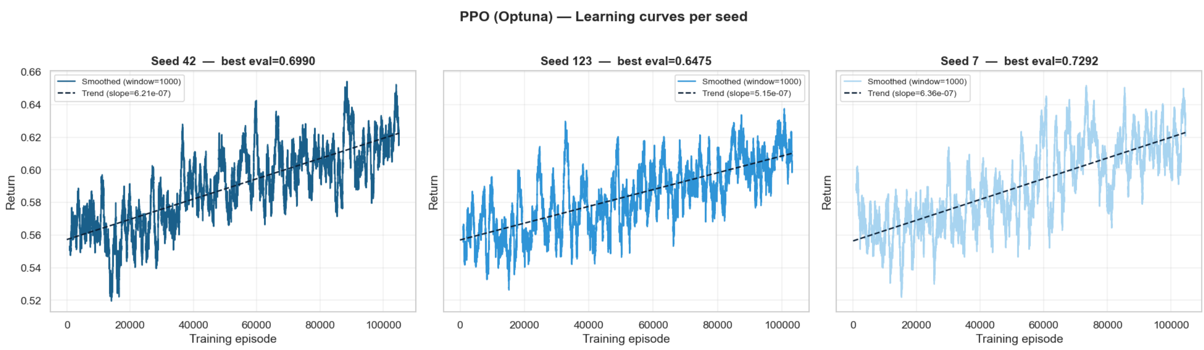

Figure 14. PPO (Optuna) learning curves across three seeds.

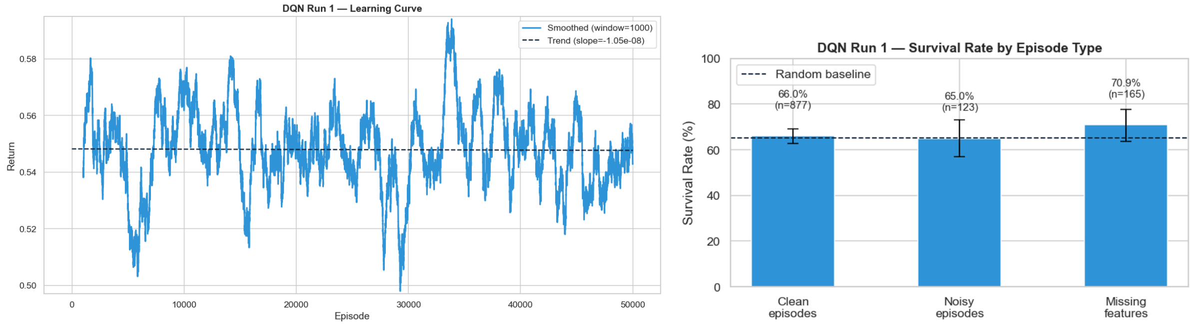

Figure 15. DQN Run 1: learning curve (left) and survival rate by episode type (right).

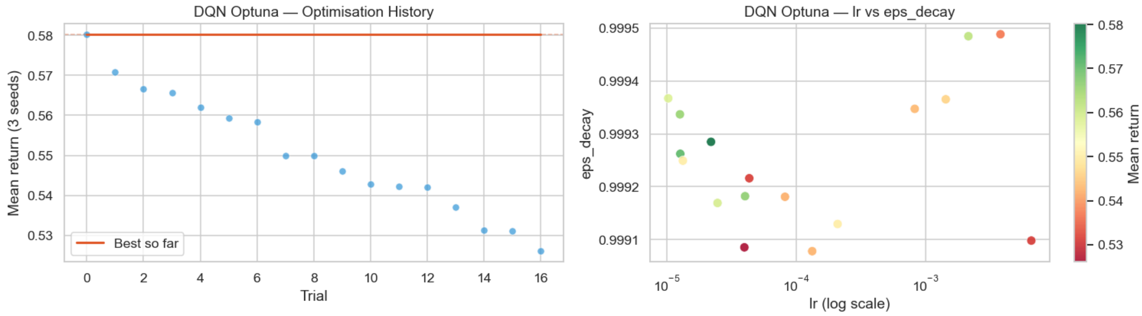

Figure 16. DQN Optuna optimisation history (left) and lr vs eps_decay scatter plot coloured by mean return (right). The best configuration was found at trial 0.

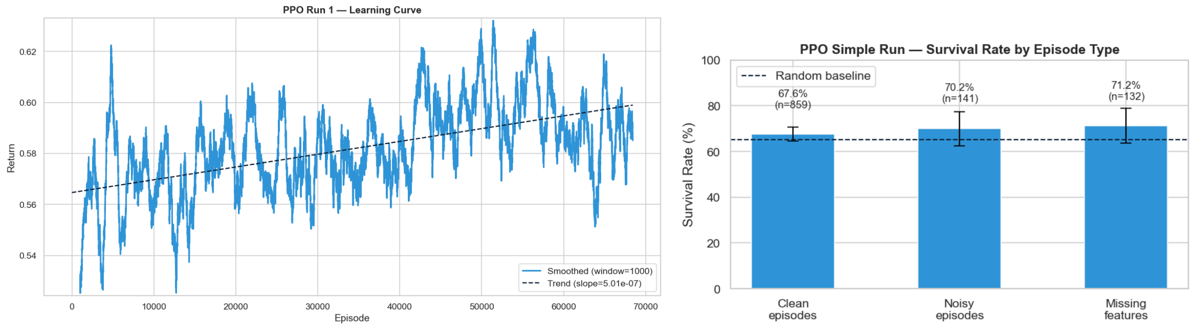

Figure 17. PPO Run 1: learning curve (left) and survival rate by episode type (right).

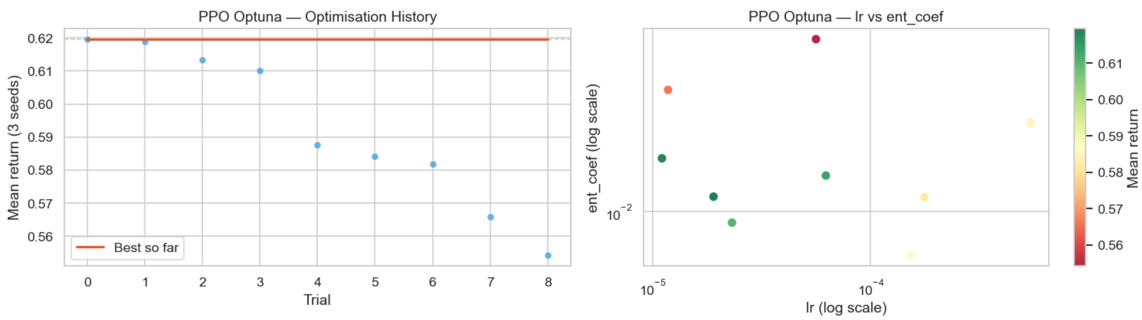

Figure 18. PPO Optuna optimisation history (left) and lr vs ent_coef scatter plot coloured by mean return (right). Low learning rates consistently dominate the top configurations.

Table 7. Mean treatment intensity by V_π quartile for each algorithm.

Quartile	PI	Dyna-Q+PS	Q-Learning
Q1 Critical	0.148	0.166	0.395
Q2 Severe	0.210	0.211	0.403
Q3 Moderate	0.177	0.186	0.328
Q4 Stable	0.108	0.111	0.320

Table 8. PPO ablation across all 2^3 wrapper combinations, averaged over three seeds.

Condition	Survival	Std	vs Baseline
Baseline (clean)	68.2%	0.7 pp	—
Noisy only	68.8%	0.9 pp	+0.6 pp
Missing only	67.4%	0.4 pp	−0.8 pp
Acute only	62.7%	1.1 pp	−5.4 pp
Noisy + Missing	68.2%	0.9 pp	+0.0 pp
Noisy + Acute	63.1%	2.2 pp	−5.1 pp
Missing + Acute	62.4%	1.9 pp	−5.8 pp
Full clinical	64.0%	2.1 pp	−4.2 pp

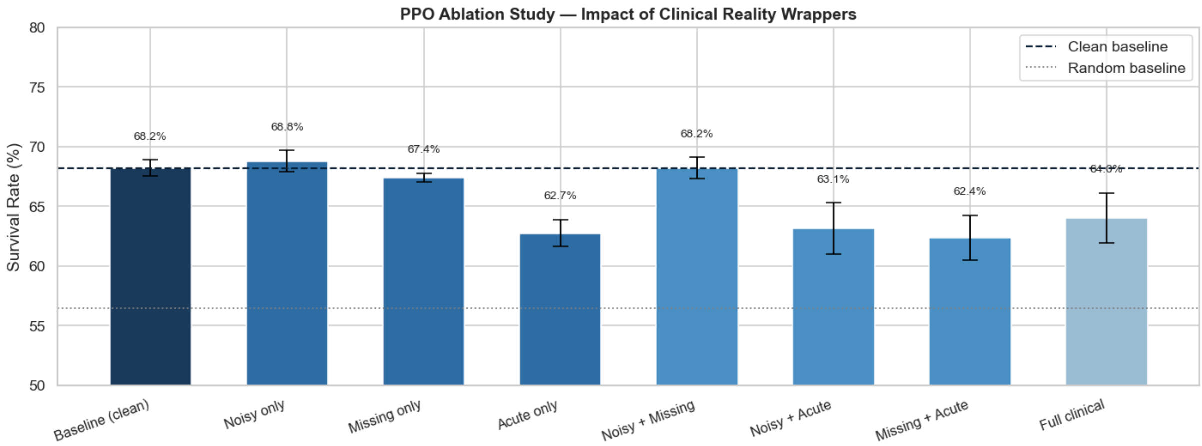

Figure 19. PPO ablation study across all $2^3 = 8$ wrapper combinations, averaged over three seeds $\{42, 123, 7\}$. Error bars show standard deviation across seeds. Colour intensity encodes the number of active wrappers (darkest = clean baseline, lightest = full clinical). The dashed line marks the clean baseline (68.2%); the dotted line marks the random policy baseline (56.4%).